

Q1. What will be the output of the following code?

```
const add = (a, b) => a + b;  
console.log(add(2, 3));
```

A) 23

B) a + b

C) 5

D) NaN

Q2. Which method returns the first element that satisfies a condition?

A) filter()

B) find()

C) map()

D) reduce()

Q3. What does the filter() method return?

A) A single value

B) A boolean

C) A new array

D) An object

Q4. Which of the following is not an ES6 feature?

A) Arrow Functions

B) let and const

C) var

D) Template Literals

Q5. Which method is used to transform each element of an array?

A) filter()

B) map()

C) find()

D) reduce()

Q6. What is the output?

```
const nums = [1, 2, 3];  
const sum = nums.reduce((acc, val) => acc + val, 0);  
console.log(sum);
```

A) 123

B) 6

C) NaN

D) undefined

Q7. How do you declare a constant in ES6?

A) var

B) let

C) const

D) static

Q8. Which method can help check if a string is a palindrome?

A) reverse()

B) sort()

C) push()

D) join()

Q9. A prime number is:

A) Divisible by 2 only

B) Has only two divisors: 1 and itself

C) Always odd

D) Has three or more divisors

Q10. Which ES6 feature lets you define a function without using the function keyword?

A) Arrow Functions

B) Classes

C) Hoisting

D) Destructuring

Q1. Define an arrow function. How is it different from a regular function?

Ans) Arrow functions were introduced in ES6. And it introduced a simple and shorter way to create functions.

Difference b/w arrow and regular function :

(i)(a) Regular function ::

```
function add(a, b) {
```

```
  return a + b;
```

```
}
```

(b) arrow function ::

```
const add = (a, b) => a + b;
```

(ii) Arrow function does not support this keyword. But regular function support this keyword.

(iii) Arrow function does not support new keyword . But regular function support.

(iv) Arrow function does not support hosting feature . But regular function support hosting feature.

Q2. What is the use of map() in JavaScript? Write a short example.

ans) map() is function not a statement . map() creates a new array by performing some operations on each array element.

Example :

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>map() Example</title>

</head>

<body>

  <script>

    const numbers = [1, 2, 3, 4, 5];

    const squaredNumbers = numbers.map(num => num * num);

    console.log(squaredNumbers);

  </script>

</body>

</html>
```

Output: [1, 4, 9, 16, 25]

Q3. Differentiate between filter() and find() with examples.

filter() : filter an array with values that passes a test . It ceates a new array.

Find() : find used to find out our first matched condition data.It always return a single object.

find example :

```
var a=[11,22,33,44];
var even = a.find((x)=>{
  if(x%2==0)
    return x;
});
console.log(even);
```

output : 22

filter example :

```
<script>
const numbers = [5, 12, 8, 130, 44] ;
const filteredNumbers = numbers.filter(function(number) {
  return number > 10;
});
console.log(filteredNumbers);
</script>
```

Output: [12, 130, 44]

Q4. Explain the use of reduce() with an example to sum numbers in an array.

Reduce(): reduce() reduces an array in a single value.

Example :

```
<script>
const numbers = [1, 2, 3, 4, 5];
const sum = numbers.reduce(function(accumulator, currentValue) {
```

```
    return accumulator + currentValue;
  }, 0);
console.log(sum);
</script>
```

Output: 15

Section C

Q1. Write a function using arrow syntax to check if a given string is a palindrome.

Input: "level"

Output: true

```
<script>

const isPalindrome = (str) => {
  const reversed = str.split('').reverse().join('');
  return str === reversed;
};

console.log(isPalindrome("level"));

</script>
```

Output: true

Q2. Use filter() and map() to return squares of all odd numbers from a given array.

Input: [1, 2, 3, 4, 5]

Output: [1, 9, 25]

```
<script>

const getOddSquares = (arr) => {
  return arr.filter(num => num % 2 !== 0).map(num => num * num);
};

console.log(getOddSquares([1, 2, 3, 4, 5]));

</script>
```

Output: [1, 9, 25]

Q3. Write a function using arrow syntax to check if a number is prime.

Input: 7

Output: true

```
<script>

const isPrime = (num) => {

  if (num <= 1) return false;

  for (let i = 2; i <= Math.sqrt(num); i++) {

    if (num % i === 0) return false;

  }

  return true;

};

console.log(isPrime(7));

</script>
```

Output: true

Q4. Given an array of strings, use reduce() to concatenate them into a single string with hyphens.

Input: ["Hello", "World", "JS"]

Output: "Hello-World-JS"

```
<script>

const strings = ["Hello", "World", "JS"];

const concatenatedString = strings.reduce((accumulator, currentValue) => {

  return accumulator + '-' + currentValue;

});

console.log(concatenatedString);

</script>
```

Output: "Hello-World-JS"

Q5. API URL : <https://medicinerestapi2.glitch.me/api/medicines/all>

Request

Type : GET

Fetch all

medicines and display on frontend in table format.

Note :

should display medicine_image in table also.

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Medicines List</title>
```

```
<style>
```

```
table {
```

```
width: 100%;
```

```
border-collapse: collapse;
```

```
}
```

```
th, td {
```

```
border: 1px solid #ddd;
```

```
padding: 8px;
```

```
text-align: center;
```

```
}
```

```
th {
```

```
background-color: #f4f4f4;
```

```
}
```

```
img {
```

```
max-width: 100px;
```

```
height: auto;
```

```
}
```



```
</style>

</head>

<body>

  <h1>Medicines List</h1>

  <table>

    <thead>

      <tr>

        <th>Medicine ID</th>

        <th>Medicine Name</th>

        <th>Composition</th>

        <th>Price</th>

        <th>Quantity</th>

        <th>Image</th>

        <th>Entry Date</th>

      </tr>

    </thead>

    <tbody id="medicine-table-body">

    </tbody>

  </table>

  <script>

    async function fetchMedicines() {

      try {

        const response = await
fetch('https://medicinerestapi2.glitch.me/api/medicines/all');

        const medicines = await response.json();

        const tableBody = document.getElementById('medicine-table-body');

        medicines.forEach(medicine => {

          const row = document.createElement('tr');
```

```

row.innerHTML = `
    <td>${medicine.medicine_id}</td>
    <td>${medicine.medicine_name}</td>
    <td>${medicine.medicine_comp}</td>
    <td>${medicine.medicine_price}</td>
    <td>${medicine.medicine_qty}</td>
    <td></td>
    <td>${new Date(medicine.medicine_entry_date).toLocaleDateString()}</td>
`;

tbody.appendChild(row);
});
} catch (error) {
    console.error('Error fetching medicines:', error);
}
}

fetchMedicines();
</script>
</body>
</html>

```



Medicines List

Medicine ID	Medicine Name	Composition	Price	Quantity	Image	Entry Date
med-82152	Azax -500 Tablets	Azithromycin Tablets	222	20		9/4/2025
med-34563	Calpol 650	Paracetamol	10	10		9/4/2025
med-27682	Pan-40	Pantaprazol tablets	250	10		10/4/2025
med-34549	Crocin Paracetamol	Paracetamol Tablets	25	10		10/4/2025
med-69248	Disprin Plus	Paracetamol Tablets, Painkiller,Pain Reliever	10	2		10/4/2025